

RA Work on Projects related to NYC EMS(Emergency Medical Service) Data

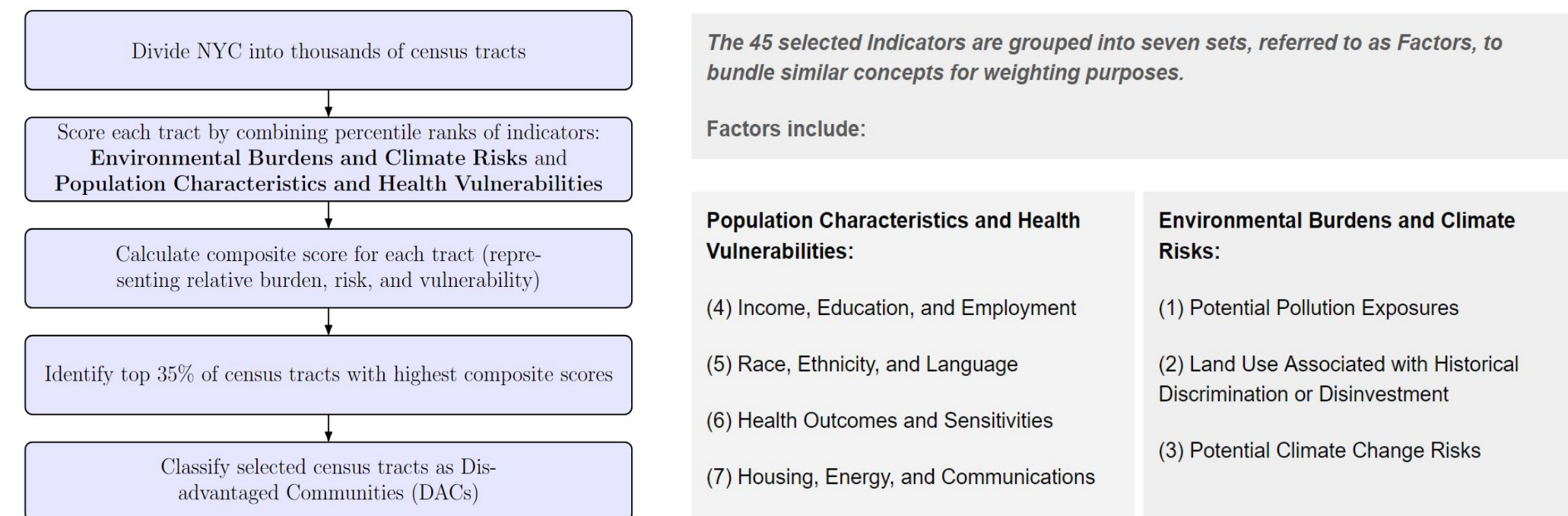
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Alternative DAC Classification

Overview

- **DAC:Disadvantaged Communities**
- Proposed and calculated an **alternative** DAC classification method.
- Integrated multiple ACS **demographic** and **socioeconomic** variables (education, public assistance, occupation, transit use) at the zip-code level, weighted by population.
- Compared the new method against traditional income/race-based DAC classification to better reflect comprehensive community vulnerability.

How DAC is Identified?



Composite Score:

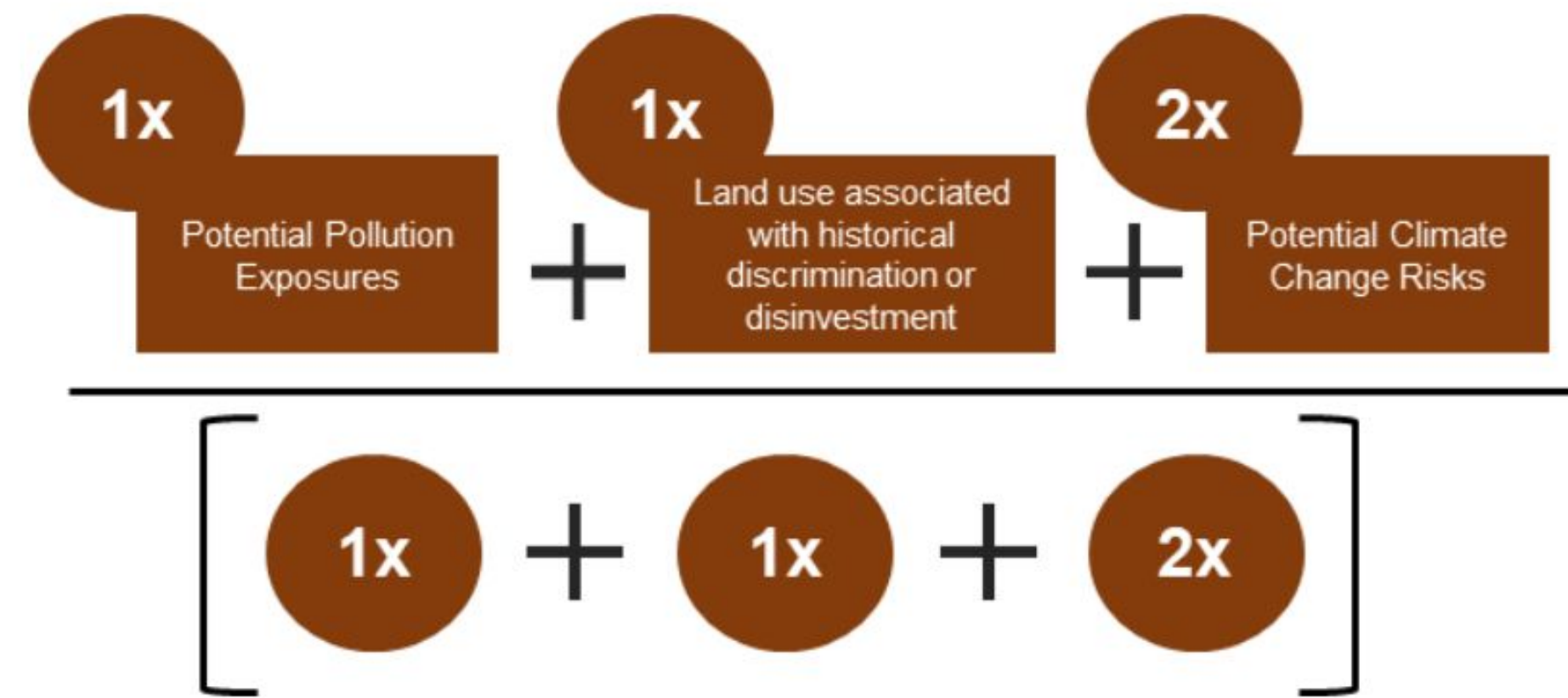


Fig 1: DAC Composite Score Calculation in Environmental Burdens and Climate Risks

DAC in Zip Code Level

- Calculated DAC scores by integrating tract-level scores with population weights, resulting in an aggregated zip-code-level DAC score .

$$DAC_{zipcode} = \frac{\sum(population_{tract} \times DAC_{tract})}{\sum population_{tract}}$$

Recovered Dispatch Zones

Overview

- **FDNY does not publicly disclose** official EMS dispatch zone boundaries.
- **Inferred and reconstructed** dispatch zone boundaries using historical EMS dispatch data (2013-2018).
- Enabled detailed analyses of **how dispatch services differ across NYC neighborhoods**.

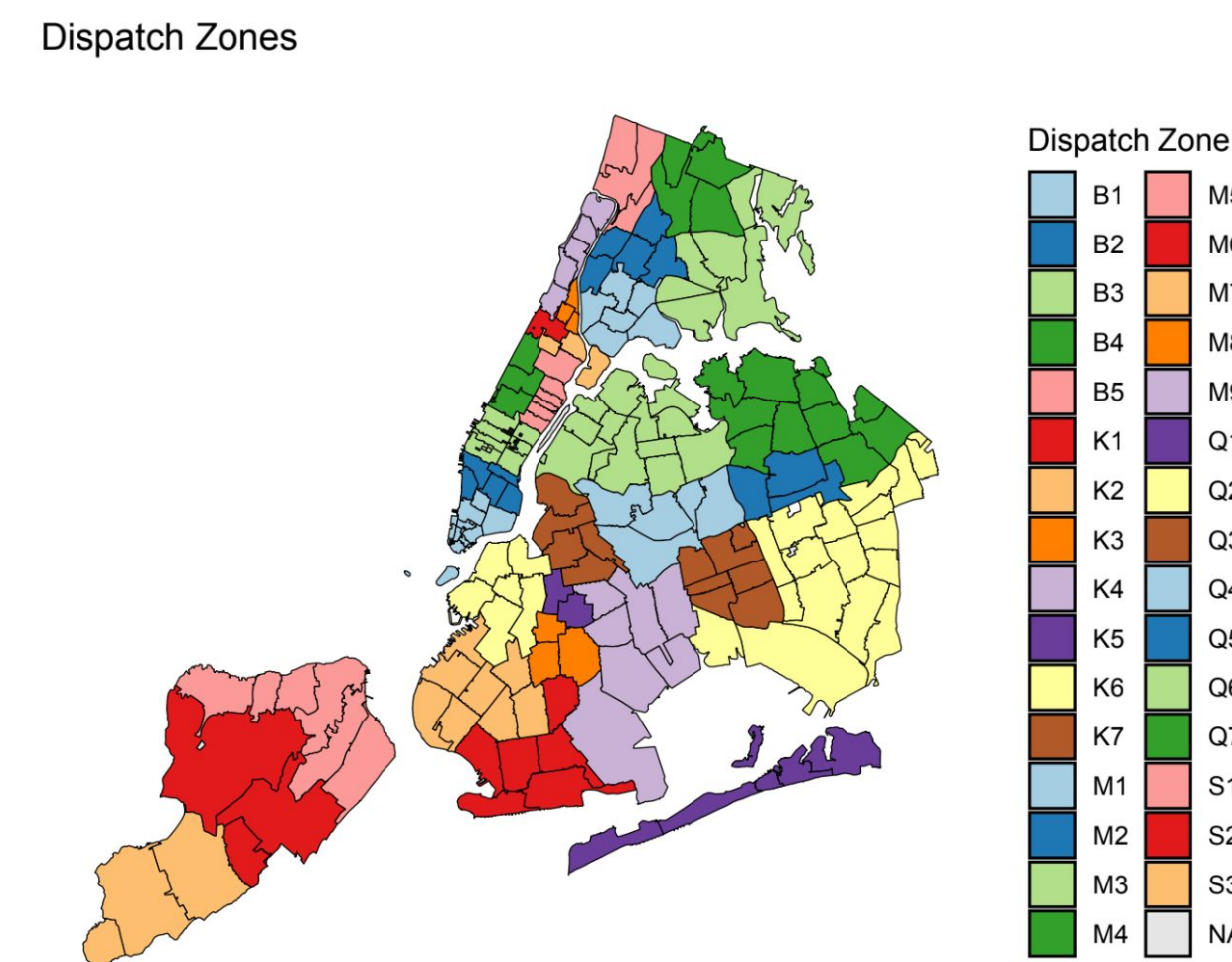


Fig 2: Dispatch Area Map

Dispatch Analysis

Overview

- Disadvantaged (DAC) areas exhibit significantly **lower EMS dispatch concentration (HHI)**, suggesting these communities rely on a more diverse set of ambulance providers.
 - During **peak emergency hours**, DAC receive proportionally more support from **secondary (non-primary) dispatch areas**, though overall response time differences remain minimal compared to non-DAC areas.
 - Strong correlations exist between community socioeconomic characteristics and EMS dispatch patterns, indicating socioeconomic factors influence ambulance dispatch allocation.
- Correlation Matrix for ACS Data

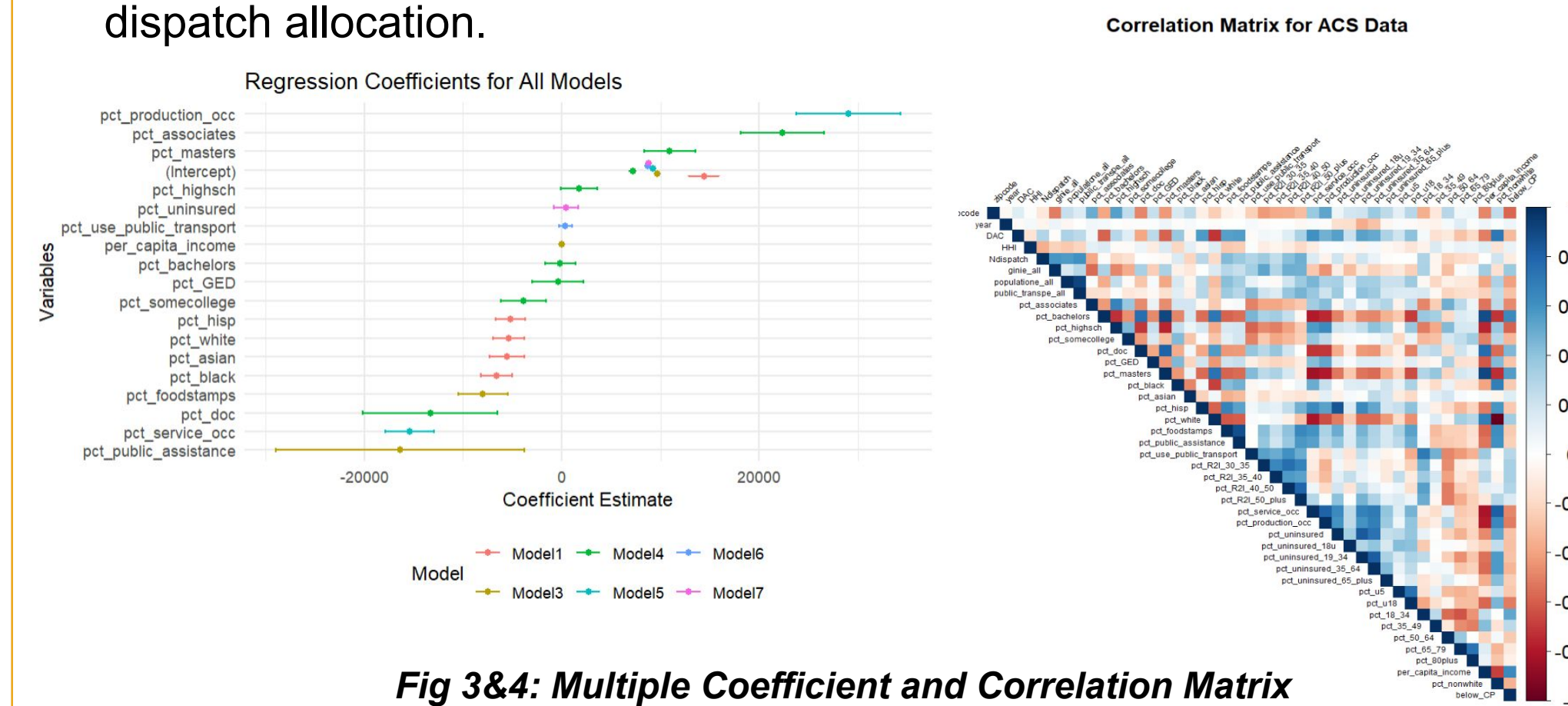


Fig 3&4: Multiple Coefficient and Correlation Matrix

Healthcare Utilization and Cost Disparities: by DAC

Overview

- **Analyzed key economic health outcomes**—including length of stay, total charges, and insurance/demographic mix—stratified by DAC status for respiratory, asthma, and injury patient groups.

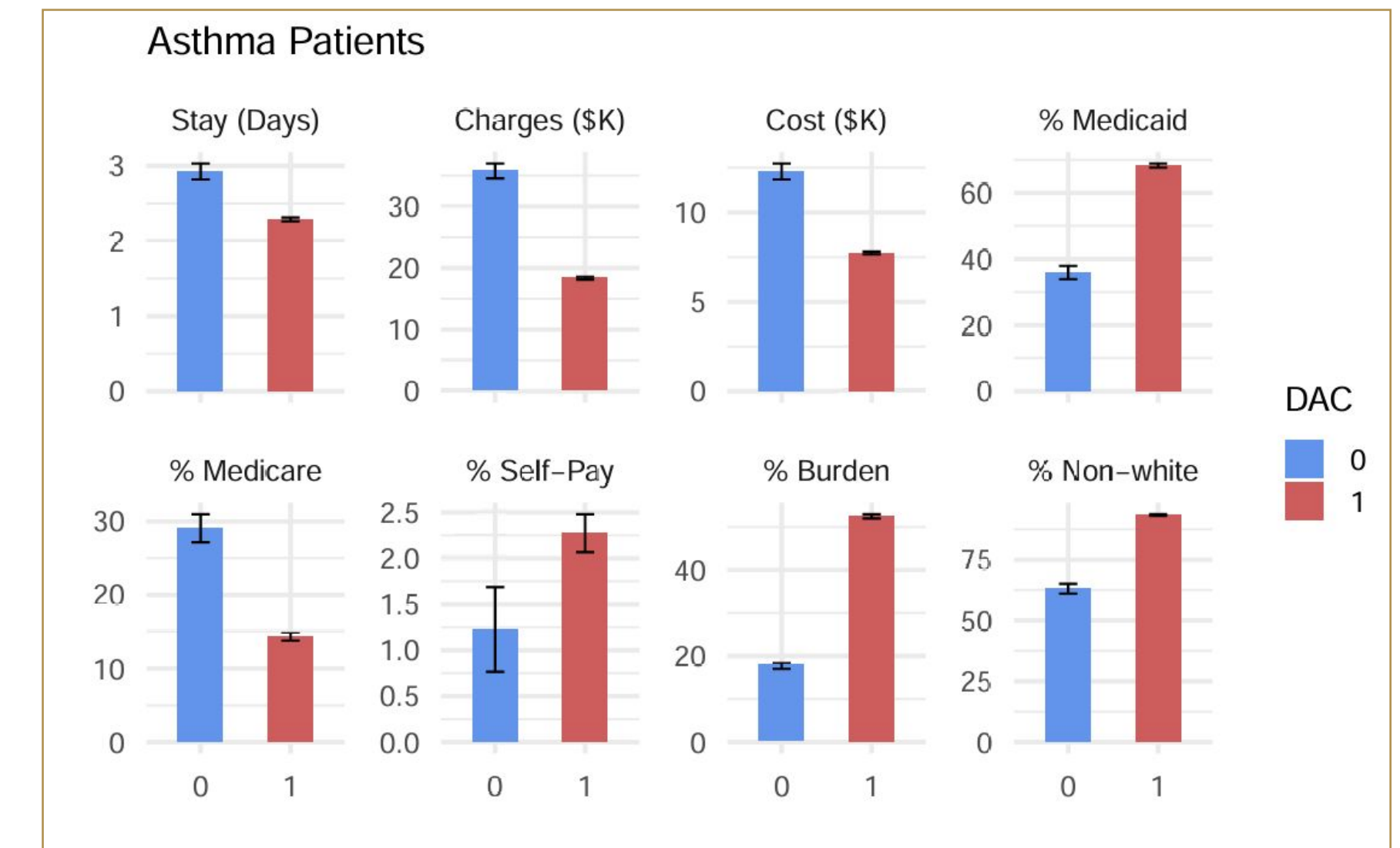


Fig 5: Bar Chart of DAC Differences in Asthma Patients

Other Works

1. Built a **web-scrapers** to collect and verify zip codes of hospitals throughout NYC, facilitating accurate geographical analyses.
2. Created a detailed **bibliography** and managed **references** for comprehensive project documentation.
3. Generated structured **LaTeX tables** for efficient and clear data presentation.
4. Performed systematic **data cleaning and preprocessing** of EMS dispatch records for robust analytical quality.
5. Supported exploratory **data visualization tasks** for internal validation and preliminary analysis.

[illegible]

Notes: Detailed descriptions of case codes can be found [here](#). Asthma calls (ASTHFC and ASTHFT codes) are included within the respiratory category for general analyses, but these cases are analyzed separately when examining asthma-specific outcomes.

```

import requests
from test import BeautifulSoup

import sys
import re

def scrape_hospitals(url, output_csv="my_hospitals.csv"):
    """
    # 1. Fetch the page
    response = requests.get(url)
    response.raise_for_status()

    except requests.exceptions.RequestException as e:
        print("Error fetching the webpage: {}".format(e))
        return

    # 2. Parse the HTML
    soup = BeautifulSoup(response.text, features="html.parser")

    # 3. Extract raw text
    # Using separator='\n' ensures each HTML block is separated by newlines in the output.
    full_text = soup.get_text(separator='\n')

    # 4. Split into lines, stripping whitespace and ignoring blank lines
    lines = [line.strip() for line in full_text.split('\n') if line.strip()]

```

Fig 6&7: LaTeX Tables and Web-Scraper Scripts in Python